

EXPERIMENT-31

Algorithm:

1. Start.
2. Declare the 2D array a, row-sum array rowSum, and required variables.
3. Input the number of rows and columns.
4. Check whether the matrix dimensions are valid.
5. Input all elements of the matrix using nested for loops.
6. Display the menu repeatedly using a do-while loop.
7. Input the user's choice and use switch:
Choice 1: Traverse the complete matrix and add all elements to sum.

Choice 2: Calculate the sum of every row and store each result in rowSum[].

Choice 3: Print a[j][i] instead of a[i][j] to display the transpose.

Choice 4: Exit the program.

Otherwise, display an invalid-choice message.
8. Continue until the user selects 4.
9. Stop.

Test Case:

For:

Rows = 2

Columns = 3

Matrix:

1 2 3

4 5 6

Choice 1:

Sum of all elements = 21

Choice 2:

Row sums:

Sum of row 1 = 6

Sum of row 2 = 15

The row-sum array contains: [6, 15]

Choice 3:

Transpose of the matrix:

1 4

2 5

3 6

The important concept here is that the original matrix is 2×3 , while its transpose becomes 3×2 .

Program, Compilation and Execution:

```
#include <stdio.h>

int main()
{
    int a[10][10], rowSum[10];
    int rows, cols;
    int i, j, choice;
    int sum;

    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    if (rows <= 0 || cols <= 0 || rows > 10 || cols > 10)
    {
        printf("Invalid matrix size!\n");
        return 0;
    }

    printf("Enter the matrix elements:\n");

    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < cols; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    do
    {
        printf("\n—— MENU ——\n");
        printf("1. Find sum of all elements\n");
        printf("2. Find sum of each row\n");
        printf("3. Find transpose of matrix\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                sum = 0;

                for (i = 0; i < rows; i++)
                {
                    for (j = 0; j < cols; j++)
```

```
        {
            sum = sum + a[i][j];
        }
    }

    printf("Sum of all elements = %d\n", sum);
    break;

case 2:
    printf("Row sums:\n");

    for (i = 0; i < rows; i++)
    {
        rowSum[i] = 0;

        for (j = 0; j < cols; j++)
        {
            rowSum[i] = rowSum[i] + a[i][j];
        }

        printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
    }

    printf("The row-sum array contains: [");

    for (i = 0; i < rows; i++)
    {
        printf("%d", rowSum[i]);

        if (i < rows - 1)
        {
            printf(", ");
        }
    }

    printf("]\n");
    break;

case 3:
    printf("Transpose of the matrix:\n");

    for (j = 0; j < cols; j++)
    {
        for (i = 0; i < rows; i++)
        {
            printf("%d\t", a[i][j]);
        }

        printf("\n");
    }
    break;

case 4:
    printf("Exiting the program...\n");
    break;

default:
    printf("Invalid choice! Please enter a choice from 1 to 4.\n");
}

} while (choice != 4);

return 0;
}
```

```
kali@kali: ~  
Session Actions Edit View Help  
└─$ ./31  
Enter the number of rows and columns: 2  
3  
Enter the matrix elements:  
1  
2 3 4  
5 6 7  
  
—— MENU ——  
1. Find sum of all elements  
2. Find sum of each row  
3. Find transpose of matrix  
4. Exit  
Enter your choice: Invalid choice! Please enter a choice from 1 to 4.  
  
—— MENU ——  
1. Find sum of all elements  
2. Find sum of each row  
3. Find transpose of matrix  
4. Exit  
Enter your choice: 1  
Sum of all elements = 21  
  
—— MENU ——  
1. Find sum of all elements  
2. Find sum of each row  
3. Find transpose of matrix  
4. Exit  
Enter your choice: 2  
Row sums:  
Sum of row 1 = 6  
Sum of row 2 = 15  
The row-sum array contains: [6, 15]  
  
—— MENU ——  
1. Find sum of all elements  
2. Find sum of each row  
3. Find transpose of matrix  
4. Exit  
Enter your choice: 3  
Transpose of the matrix:  
1 4  
2 5  
3 6  
  
—— MENU ——  
1. Find sum of all elements  
2. Find sum of each row  
3. Find transpose of matrix  
4. Exit  
Enter your choice: █
```

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